24-case-studies

Stat 230 - Spring 2026

1 Common issues on case studies

- Correct things in an incorrect order
- What goes in an EDA?
- What types of plots go in the main text?
- Report appendix vs quarto appendix

2 General order

- Intro:
  - What is the research question framed in general? What background information should the reader have to understand your study? How does this research question translate to a question we can answer with a statistical model? What is the response and (general) predictor variables?
- EDA/Data:
  - What dataset are you using? How many observations/variables? What is the distribution of the response variable? How are the predictors related to the response variable? Is there collinearity or skewness that should be addressed before fitting any models?
- Results:
  (a) Model building: what were your candidate models? what were your criteria for choosing a model? (You can report *theoretical* model here)

(b) Conditions and diagnostics: does everything look OK? Did you need to do any further investigation?
(c) Now that you have verified the validity of your model, you can report the *fitted* model equation or the coefficient table
(d) Interpretations of coefficients (with uncertainty!) or results from tests. Answer your research question here

- Discussion:
  - Reiterate your main result without any technical results
  - Are there any limitations to your analysis?
  - If you had another week, what could you investigate? If you had another dataset, what could you investigate?
- Report appendix
  - These are graphs that you *directly refer to in the text* but that *aren't central for understanding the results*.
    * "We investigated scatterplots of all predictors against each other to assess collinearity. Variable x_1 and x_2 were highly correlated ($r = .99$), and so decided to drop x_2 from the model." – the reader does not need to see the pairs plot in the main text to understand what you did, but may want to skim through to see what you saw.
  - You may also want to include model statements for models you tested but don't necessarily want to talk about:
    * "We tested 10 different models (see appendix for model statements) and chose the following based on Adjusted R^2 and BIC:"
- Quarto appendix
  - This is your code for all analyses (audience is me)
  - Do not assume the reader has access to this appendix

3 Presenting equation

My recommendation is to report the *theoretical* equation first, then include a coefficient table:

After determining that no transformations were needed, the binomial model with an interaction term is given by:

,

$$, \log\left[\frac{P(\text{Removed})}{1-P(\text{Removed})}\right] = \beta_0+\beta_1(\text{morph}_{\text{light}})+\beta_2(\text{distance})+\beta_3(\text{morph}_{\text{light}}\times\text{distance}), \tag{1}$$

Based on binned residual plots and a ad-hoc test of overdispersion ($\psi \approx 1$), we determine that binomial logistic regression conditions are satisfied. The estimated coefficients with 90% confidence intervals are given in Table 1.

```
broom::tidy(moth_glm, conf.int = TRUE, conf.level = .9) |>
  select(-c("statistic", "p.value")) |>
  knitr::kable(digits = 2)
```

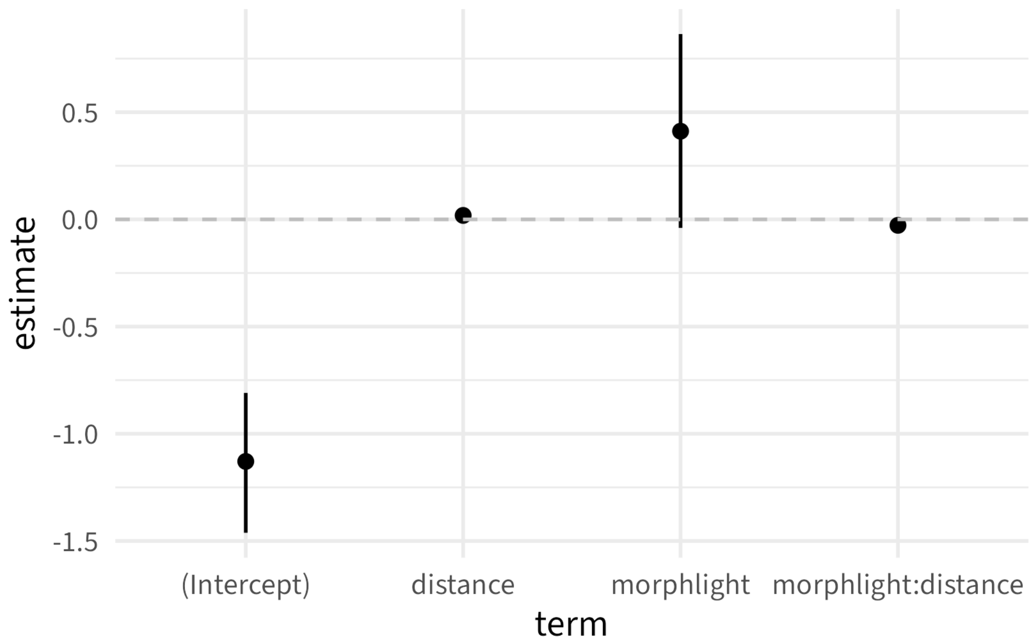
Table 1: Coefficient from moth binomial regression model.

term	estimate	std.error	conf.low	conf.high
(Intercept)	-1.13	0.20	-1.46	-0.81
morphlight	0.41	0.27	-0.04	0.86
distance	0.02	0.01	0.01	0.03
morphlight:distance	-0.03	0.01	-0.04	-0.01

4 Coefficient graph

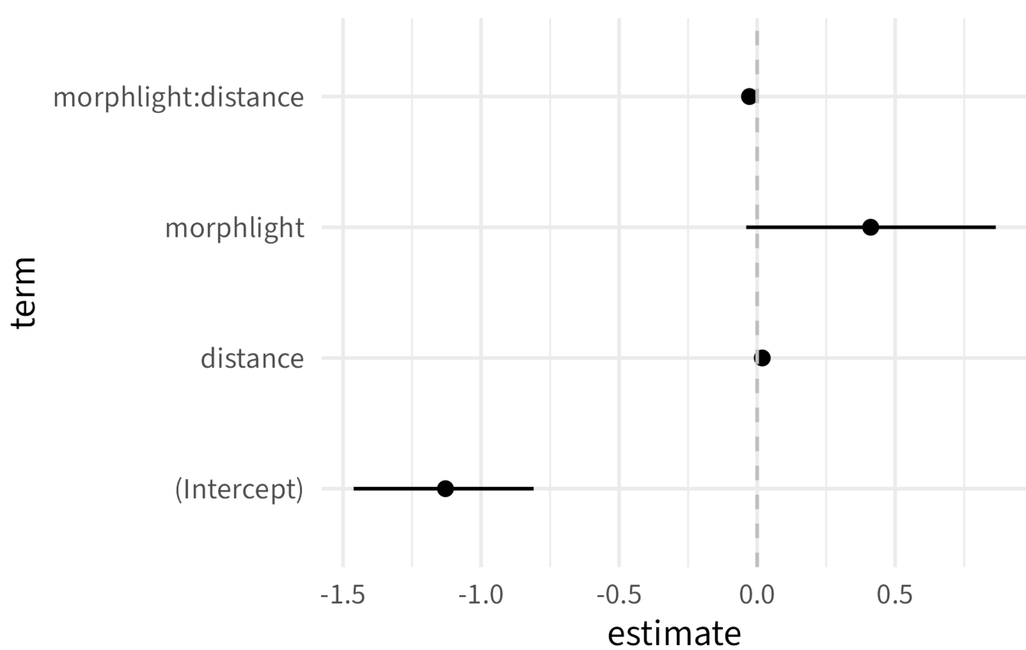
Alternatively, put coefficient table in the report appendix and include a coefficent graph

```
broom::tidy(moth_glm, conf.int = TRUE, conf.level = .9) |>
  gf_pointrange(estimate + conf.low + conf.high ~ term) |>
  gf_hline(yintercept = 0, linetype = "dashed", col = "gray")
```



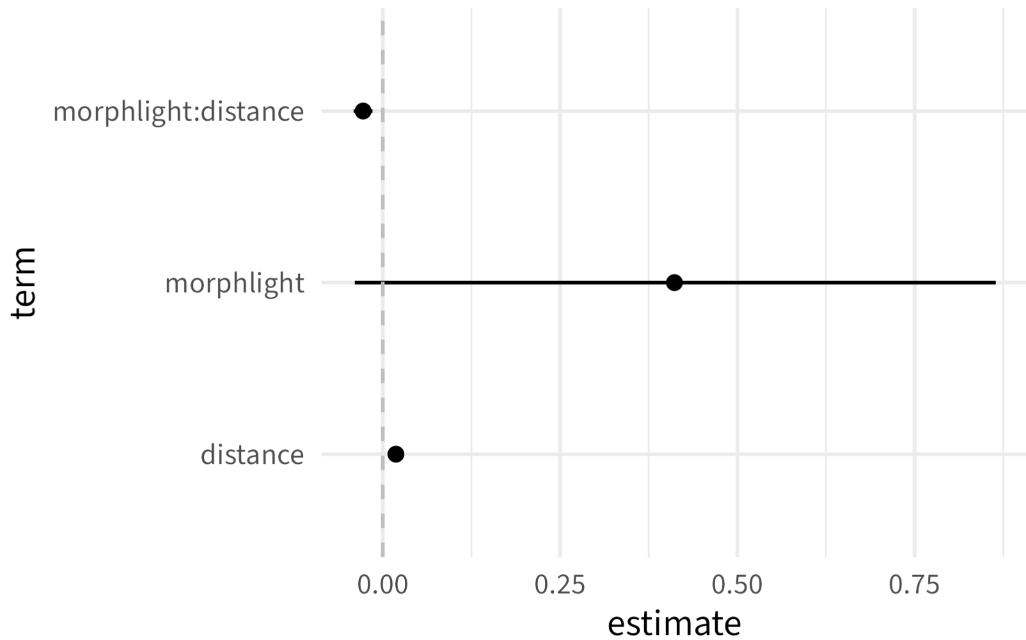
I prefer to flip the coordinates here so that the x axis labels don't overlap:

```
broom::tidy(moth_glm, conf.int = TRUE, conf.level = .9) |>
  gf_pointrange(estimate + conf.low + conf.high ~ term) |>
  gf_hline(yintercept = 0, linetype = "dashed", col = "gray") +
  coord_flip()
```



Removing intercept can often make the remaining terms easier to read:

```
broom::tidy(moth_glm, conf.int = TRUE, conf.level = .9) |>
  filter(term != "(Intercept)") |>
  gf_pointrange(estimate + conf.low + conf.high ~ term) |>
  gf_hline(yintercept = 0, linetype = "dashed", col = "gray") +
  coord_flip()
```



5 Examples:

- Multiple linear regression example)
- Logistic regression example
- Advanced method example
- Experiment example
- A logistic regression example